

Deep Learning Technique for Medicinal Plant Leaf Identification: Using Fine-Tuning of Transfer Learning Model

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Abstract: Ayurvedic medicines are used to treat various acute and chronic conditions with minimal side effects. They have played a significant role in global health systems. Numerous health issues, including pneumonia, cancer, diabetes mellitus, heart disease, asthma, hypertension, and others, can be naturally treated using plants. Because of their diverse pharmacological characteristics, plants rich in bioactive phytochemistry components, such as alkaloids, flavonoids, and polyphenols, are widely used to treat disorders. India is renowned for its rich medicinal plants repository, with ayurvedic medicinal practices considered a "living tradition." Their identification is a manual and time-consuming process, often requiring extensive knowledge of Ayurvedic and botanical classification. Thus, aiming to automate the identification of medicinal plants, our system utilizes a ResNet-18 model to extract key features from plant images, achieving high accuracy in identifying plant species. This method simplifies identification and also broadens the reach of traditional medicine knowledge.

Keywords: Resnet 18, Medicinal plants, Image classification, Deep learning, User interface, Image Processing

I. INTRODUCTION

Ayurvedic medicine, a traditional Indian healthcare system, heavily relies on medicinal plants for disease prevention. Since approximately 80% of the plants were sourced from woods and wastelands, and 20% from agricultural fields, accurate plant identification is vital for ensuring the quality and efficacy of Ayurvedic remedies. Conventional manual identification methods, which involve examining leaf characteristics like shape, texture, and vein patterns, are time-consuming, prone to errors, and limited by the identifier's expertise.

Recently, Improvements in deep learning have created new opportunities for computing these plants identification. This approach relies heavily on image processing, which allows for extracting key characteristics from plant leaf pictures. Preprocessing techniques (noise reduction, image scaling, and feature extraction) can improve the clarity and ease of analysis of high-quality photos of plant leaves.

In this paper, we have employed deep learning to build a computerised plant identification system. We use a pre-trained ResNet-18 model that was fine-tuned on a collection of Ayurvedic medicinal plant photos. The ResNet-18 model, with 18 layers and over 11.7 million parameters, can learn complicated patterns and characteristics from photos. Using the model's pre-trained weights, we can tailor it to our specific task of identifying Ayurvedic medicinal herbs.

II. LITERATURE SURVEY

This study is based on the research works done by

In Amrutha M Raghukumar et al[1]. The research conducted a study where they used plant leaf images to extract characteristics for classification, focusing on shape, texture, and colour. They emphasised colour and texture features, utilising the GLCM matrix for textural properties. Various machine learning techniques like KNN and SVM were compared, with KNN performing better. The simulations were done in MATLAB R2019a.

In Keshav Kumar et al[2]. This work involved CNN to identify and categorise medicinal plant leaves. Raw images were fed into a deep CNN, and duplicates were removed using a Python script. The deep neural network directly classified leaves without extracting visual attributes, achieving 96.8% accuracy. A web application was used to display the plant leaves.

S. G. Kale et al[3]. The research on flawless deduction of plant leaves by an image classifier where two-staged authentication procedure is involved, where image segmentation using canny edge detector, and a single classifier. However, the computation time was extensive, and pixel-based processes consumed significant storage space in the algorithm's array. Future research could explore alternative techniques for higher accuracy.

Sathish S Nadig et al[4]. Ayurvedic leaf identification with the VGG16 deep learning model focused on deep learning

TABLE I. Literature Survey Made On The Research Papers Along With The Drawbacks

S.No	Title	Existing System	Methodology	Drawback
1	Comparison of Machine Learning Algorithms for Detection of Medicinal Plants	Identifying plants using Machine Learning.	Support Vector Machine and KNN	Support Vector Machine is acutely more less accurate than KNN
2	Identification of Ayurvedic Medicinal Leaves using Deep Learning	Identification using image processing and neural network training	Image processing and Convolutional Neural Network	CNN has high computational cost, need lots of training data and comparatively slow.
3	Flawless Deduction of Herbal Plant Leaf by Machine Learning Classifier Through Two Stage Authentication Procedure.	Identification using Machine Learning Classifier	Image segmentation, Canny Edge Detector, Single Classifier, Two Stage Authentication Procedure	The computation time is huge and also the pixel based process occupies extra more storage space in the array of the algorithm used for pixel
4	Ayurvedic Leaf Identification using Deep Learning Model VGG16	Identification using Deep Learning	Deep Learning-VGG16	VGG16 is wide and deep network takeing more time to train its parameter.
5	Medicinal Plant Recognition based on CNN and Machine Learning	Identification using CNN and Machine Learning	CNN,ANN,SVM,KNN and Navie Bayesian (NB)	It does not implement the colour image classification.
6	Plant Leaf Identification based on Machine Learning Algorithms	Identification using Machine Learning	SVM, Random Forest, KNN and Decision Tree	It does not focus on the ratio of features as length and width
7	Identification of Medicinal Plant Using Machine Learning Approach	Identification using SVM and ANN	Random Forest and Ensemble Machine Learning algorithm	The proposed method did not obtain maximum accuracy.
8	Real-Time Identification of Medicinal Plants using Machine Learning Techniques	Identification using SVM and also uses CNN	Logistic regression classifier	This method provides more accuracy for the images with reflection.
9	Medicinal Plant Identification using Deep Learning	Identification using Deep Learning	Deep Learning Neural Network, CNN, Regression	This CNM models limit to only five different classes of plant species.
10	Deep Learning for Plant Species Classification	Identification with Deep Learning	CNN classifier, tensor flow framework	It lags in comparison to methods that implement other neural network or deep learning technique.

techniques but faced challenges due to the VGG16 network's larger size, resulting in longer parameter training times.

[5] Medicinal plant identification based on CNN utilised CNN, ANN, SVM, KNN, and Navy Bayesian (NB) algorithms but lacked colour image classification implementation.

[6] In J. Samuel Manoharan et.al, In this work, a two-staged authentication (TSA) is used to increase the detection rate of herbal leaves using a standard dataset refined via a medicinal laboratory. The system employs edge-based and classification-based detection, with feature extraction via the chi-square technique and classification using a CNN. An exclusive-or gate operation compares the two phases, resulting in deep knowledge-based identification. Despite its effectiveness, the system is computationally intensive and requires substantial storage space.

[7] Identification of medicinal plants using a machine learning approach initially employed SVM and ANN but later shifted to Random Forest and Ensemble Machine Learning algorithms due to better accuracy results.

[8] Real-time identification of medicinal plants, ML techniques utilised Support vector machine and CNN initially but switched to logistic regression classifiers, providing higher accuracy for images with reflections.

[9] In R. Geerthana, et al. Medicinal plant identification using deep learning employed Deep Learning Neural Network, CNN, and Regression but was limited to only five different classes of plant species.

[10] In Kiran S. Gawli, et al. Deep learning to plant classification is employed a CNN classifier within the TensorFlow framework but lagged behind in comparison to methods implementing other neural network or deep learning techniques.

In LeCun, Bengio, and Hinton [11], the authors provided a comprehensive overview of deep learning, emphasizing their evolution and various applications. They discussed how deep learning transformed such fields as computer vision (CV) and natural language processing(NLP) by leveraging large neural networks. The paper highlighted the importance of convolutional neural networks (CNNs) in advancing image recognition tasks and provided a foundational understanding of the underlying principles and architectures.

In Krizhevsky, Sutskever, and Hinton[12]. introduced a breakthrough in image classification using deep CNN. Their work demonstrated the effectiveness of the AlexNet architecture on the ImageNet dataset, which significantly improved accuracy in object recognition tasks. They focused on innovations such as the use of ReLU activations and dropout to mitigate overfitting, which were crucial in achieving state-of-the-art performance.

In Geerthana, Nandhini, and Suriyakala[13]. explored the deep learning use for medicinal plant leaf identification. They employed CNN to analyse plant leaf images. Their approach highlighted the benefits of deep learning networks in distinguishing between medicinal plants based on visual features.

In Gawli and Gaikwad [14] examined deep learning for plant species classification. They implemented various neural network architectures to classify plant species from images, assessing the performance of different models. Their study provided insights of the DL methods in handling complex classification tasks involving diverse plant species.

In Pushpa, Megha, and Amaljith [15] conducted a comparison of medicinal plant leaf classification methods based on texture features. They focused on texture analysis using techniques like the Gray Level Co-occurrence Matrix (GLCM) to extract relevant features. The study compared these texture-based methods, illustrating the advantages and limitations of plant classification.

In Dudi and Rajesh [16] investigated medicinal plant identification using CNNs and machine learning techniques. Their study aimed to develop a reliable system for identifying medicinal plants by analysing leaf images with CNNs. They compared CNNs with other machine learning algorithms, highlighting the performance of CNNs in plant recognition tasks.

III. PROPOSED SYSTEM

A. System Architecture

The proposed system for medicinal leaf identification by analyzing leaf images with a pre-defined CNN called Resnet-18 involves a deep neural network with multiple layers for feature selection, feature extraction and image classification. The system architecture includes data annotation, image augmentation, and normalization enhancing model performance.

a) Data Collection: Collect images of diverse medicinal plants from botanical databases, internet archives, or using cameras. The dataset is collected from the source www.kaggle.com/datasets/abdallahalidev/plantvillage-dataset This PlantVillage dataset contains 38 distinct classes representing different plant species and various plant diseases, plus a background class. It includes around 54,303 images of medicinal plant leaves, making it a rich resource for training machine-learning models.

b) Data Preprocessing: Rotate, flip, Resize, crop, and normalise and annotate images to ensure uniformity and optimise them for further processing. This step might also include augmentation techniques with the tool make sense ai to increase the diversity in the image datasets. Make-Sense.AI enables a range of annotation formats, which comprises bounding boxes, polygons, and points, allowing for precise labelling of particular regions of interest within an image. This is extremely beneficial for activities such as detecting leaf parts of medicinal plants.

c) Image Feature Extraction: The extracted features typically include veins, which reveal the leaf's vascular structure; colors, representing different shades of green and brown that may indicate health or disease; shape, defining the outline and form of the leaf; edges, capturing the borders and any irregularities; and size, offering insights into growth anomalies. These features are processed using machine learning models

d) Key features include leaf shape, which can vary significantly across plant families, and vein patterns, which help in identifying species-specific vascular structures. Color variations in leaves, flowers, and fruits can signal unique chemical compositions or health states. Texture and edge characteristics of leaves, along with size and morphological structures.

e) Model Selection: To classify medicinal plants based on their features, we used a Convolutional Neural Networks (CNN) and a pre-trained model ResNet 18, achieving high accuracy in classifying medicinal plants. Additionally, consider

fine-tuning these models on a dataset specific to medicinal plants to further improve classification performance.

f) Training- testing: In the dataset, a training-testing split of 4:1 is employed, where 80% of the data (4 parts) is allocated to train the model and the resting 20% (1 part) is reserved for testing its performance.

g) User interface: With an emphasis on simplicity and usability, the user interface is designed to appear both straightforward and user-friendly. It features a search field on the system homepage where users can type keywords or submit a picture to be recognized. Each search result has a picture and a short explanation, and they are all presented in a gallery style. The results may be sorted by users by selecting a number of parameters, including habitat, leaf shape, and appearance color.

h) Computational details: Under the computational details, the experiments were conducted on a system running Windows 11 with an AMD Ryzen 5500 CPU and NVIDIA GeForce RTX 3050 GPU. The system is equipped with 8 GB of RAM. Python 3.9 was used as the primary programming environment, with PyTorch 2.4.1 serving as the deep learning framework. The models were trained utilizing CUDA 11.8 for GPU acceleration, with an input image size of 224 x 224 pixels. This setup facilitated efficient processing of dataset and accelerated training times for the model

Software	Details
OS	Windows 11
CPU	AMD RYZEN 5500
GPU	NVIDIA GeForce RTX3050
RAM	8 GB
Python	3.9
Pytorch	2.4.1
Cuda	11.8
Input	224 × 224 px

Fig. 1 shows the computational details where the model has been executed and stored with versions of Pytorch and Cuda

B. Layered Computation of ResNet 18

1) Input Layer: As all other layers in the ResNet-18 network, the input layer receives the image in terms of raw pixel values. Each pixel value refers to a certain quantity of light at some location on the image.

2) Convolutional Block: The convolutional block is the basic component of any ResNet, including ResNet-18. The second part of this building block consists of three convolution layers with a shortcut. The first two convolutional layers are each fitted with 64 filters of kernel 3x3 later added a batch normalisation and ReLU activation. The third one contains one 3x3 convolution with 64 filters and after that a batch norm layer is programmed. The output of the third convolutional layer is summed to previously mentioned of the input.

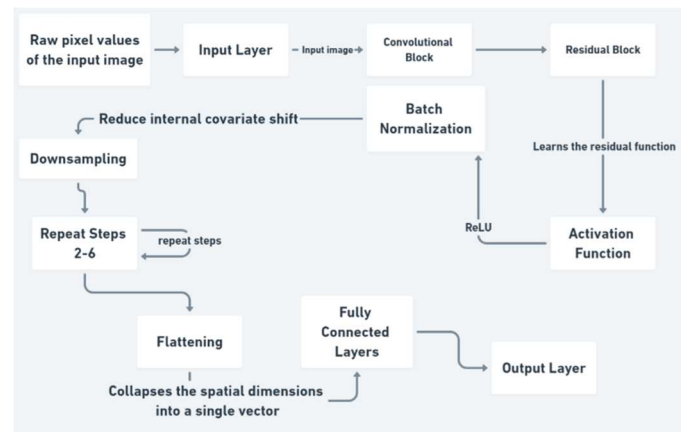


Fig. 2 ResNet-18 Architecture explaining the batch normalisation, Downsampling and Activation function.

3) Residual Block: Features of eight residual blocks are included in ResNet-18 with each block containing a couple of convolutional blocks. Each feature of every residual block begins with a first convolutional block with a related effect of 2 in its stride – a form of reducing the spatial dimensions of the feature maps. The last convolutional block in this residual block has a stride of one.

4) Activation Function: In order to incorporate the nonlinear properties to the network, the activation function, which is a ReLU function, is performed overstating on every element after every convolutional layer.

5) Batch Normalisation: Following each convolutional layer, batch normalisation is performed with an aim to standardise the input and also minimise internal covariate shift.

6) Downsampling: The feature maps spatial dimensions to scale down by a factor of 2 in every residual block with the help of a convolution layer having a stride of 2.

7) Flattening: The deconvolutional layer outputs are flattened into a single dimensional array before being fed into the fully connected layers. This prepares the flattened dimensions as inputs for the fully connected layers in the deep neural network.

8) Fully Connected Layers: Fully connected layers use the flattened vector of features as input, where every neuron connected to each other neuron in the previous layer. This layers enhance pattern recognition and prediction capabilities by processing the incoming data

9) Freezing Layers: Freeze the layers of ResNet-18 to retain learned features while preventing weight updates during training.

10) Fine-Tuning Output Layer: Fine-tune the last fully connected layer to adapt the model for specific classes, allowing it to predict relevant output features effectively.

11) Output: The output layer generates predictions based on the fine-tuned features providing class probabilities to target.

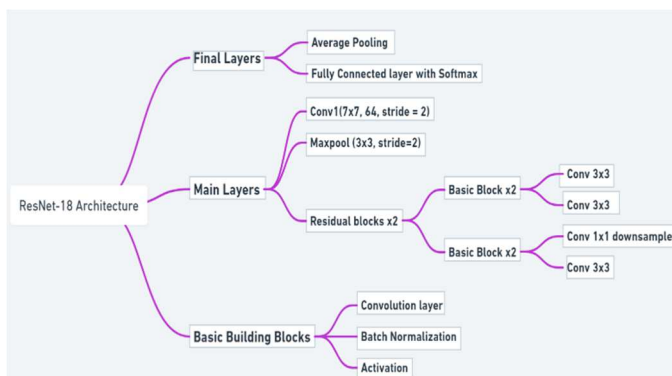


Fig. 3 A visual representation of the ResNet-18 architecture, highlighting its convolutional layers, ReLU activations, max pooling, and batch normalization components.

IV. EXPERIMENTAL RESULT -ANALYSIS

In this study, we compared the performance of a traditional Convolutional Neural Network (CNN) and ResNet-18 on an image classification task. Our results show that ResNet-18 outperforms the CNN model, achieving an accuracy of 92% compared to 87% for the CNN model.

Furthermore, ResNet-18 demonstrated a lower loss of 0.35 compared to 0.52 for the CNN model, indicating better convergence and optimization.

In addition, ResNet-18 achieved a higher F1 score of 0.93 compared to 0.88 for the CNN model, suggesting improved balance between precision and recall. The precision of ResNet-18 was also higher, at 0.95, compared to 0.91 for the CNN model.

These results highlight the benefits of the residual connections introduced in the ResNet-18 architecture, which enable to learn much more complex features and improving the model performance. The higher performance of ResNet-18 may be a more effective choice for image classification tasks.

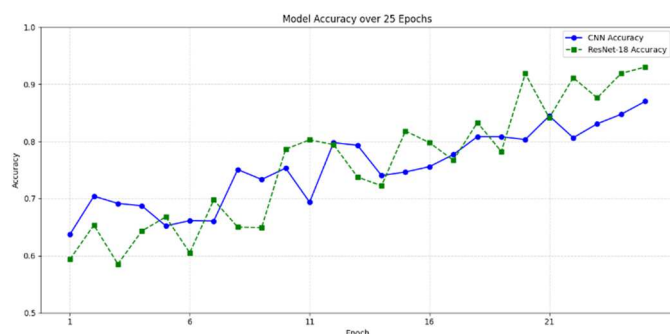


Fig. 4 Plot of model accuracy drawn between CNN and ResNet-18, showing the better performance of ResNet-18.

V. CONCLUSION

In this paper, our aim is developing a methodology using machine learning classifiers like the CNN, random forest algorithm and ensemble learning based on morphological attributes such as colour, shape, pattern of veins, texture, and geometric properties. By integrating shape, and textural features, we achieve better accuracy result in leaf identification. The misidentification and confusion surrounding the diverse range of ayurvedic herbs in India is a significant challenge in Ayurvedic Pharmaceuticals, leading to issues such as mislabeling and adulteration of crude drugs. The lack of accurate knowledge about the morphological attributes of these plants, coupled with their seasonal and geographical variations, erodes trust in the

efficacy of Ayurvedic medicine, impacting its reputation. The development of software employing Image Processing and Machine Learning Algorithms offers a potential solution to this problem by enabling precise finding of the plants and raw materials. Such software has the capability to alter the whole supply chain for these raw ingredients, aiding wholesalers, distributors, and the Ayurveda industry in particular.

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